Andy Palm

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EDUCATION

Rensselaer Polytechnic Institute (RPI) — Troy, NY
Bachelor of Science in Aerospace Engineering — *Expected May 2029*
Minor in Economics — *GPA: 3.0*

SKILLS

MATLAB/Simulink, Python, Siemens NX, SimCenter Nastran, Ansys (BladeGen, CFX, Fluent), Microsoft Excel

EXPERIENCE

Compact Operational Research Engine (CORE) — Troy, NY
Chief Engineer — *Jan 2025 – Present*

- Led a cross-functional team of 20+ students across 5 subsystems (fuel, combustion, turbomachinery, controls, safety) to design and build one of the nation's first fully student-constructed jet engines from the ground up, targeting 60,000 RPM
- Personally contributed to technical development across turbomachinery, CFD simulation, and structural analysis workstreams, including component design iteration and FEA validation on critical engine components
- Architect project-level development strategy across 5 subsystems, managing a union-funded budget and coordinating with RPI Environmental Health and Safety to ensure compliance with institutional testing requirements

NiSource — Canonsburg, PA
Engineering Intern — *May 2026 – Present*

- Developed a risk prioritization model for bridge-attached pipeline assets across 160+ culverts in 5 states, enabling data-driven repair decisions within the Distribution Integrity Management Program
- Managed a $36,000 contractor engagement to conduct field investigations of bridge-attached pipeline assets, coordinating site selection across a multi-state network and overseeing deliverable reporting through contract execution
- Built an AI-powered natural gas incident monitoring tool using Microsoft Copilot to aggregate and track pipeline safety events across national sources, enabling faster awareness of industry-wide risk patterns for the Distribution Asset and Risk Management team

BotsIQ — Washington, PA
Chief Engineer — *Aug 2022 – May 2024*

- Achieved the highest competition placement in school history (11th/66 schools) across PA and neighboring states, leading a team of 10 through full robot design, manufacturing, and competition
- Designed all mechanical components in Autodesk Inventor with weight, strength, and competitive performance constraints, validated through FEA simulation prior to manufacturing
- Produced all G-code and managed end-to-end manufacturing for robot components, coordinating across electrical, design, and manufacturing subteams to deliver a competition-ready system

PROJECTS

CORE Conditions & Task Tracker | HTML, Project Management

- Architected and built a real-time, multi-user web application to centralize task ownership and interface condition tracking across CORE's 5 subsystems, replacing fragmented spreadsheet-based coordination for a 20+ person team
- Applied systems engineering practices to define a shared task and interface data model spanning all 5 subsystems, strengthening project management visibility and accountability for subsystem leads

Parametric Inlet Geometry Optimizer | Python, Siemens NX

- Developed a Python automation pipeline integrating parametric scripts with Siemens NX journal files to programmatically generate and modify bellmouth inlet CAD geometry for CORE's student-built jet engine
- Eliminated manual CAD rework by automating expression updates across geometry iterations, enabling rapid exploration of the inlet design space in support of a 60,000 RPM rotor target

ACTIVITIES

RPI Men's Varsity Soccer — Aug 2025 – Present

RPI AIAA — Nov 2025 – Present